Project Management

(Open Elective)

Savitribai Phule Pune University		
OEL-221A : Project Management		
Teaching /scheme	Credits	Examination Scheme
Theory : 02 Hours/Week	02	CCE : 15 Marks **End-Semester:** 35 Marks

Course Objectives:

Students will be familiarized with

1. Fundamental principles of project management
2. Project planning, organizing, and controlling the projects
3. Skills in project scheduling, budgeting, and resource allocation
4. Risk management, quality control, and stakeholder management in projects
5. Project management concepts to real-world scenarios.

Course Outcomes:

- CO1: Explain the principles of project management
- CO2: Use project management concepts in real-world scenarios
- CO3: Apply Agile Project Management
- CO4: Discuss the importance of risk management, quality control, and stakeholder management in projects
- CO5: Demonstrate skills in project planning, execution, and control

Unit I -Introduction to Project Management

Course Contents
Project Definition, Project Life Cycle, processes and Knowledge areas in Project management, WBS and its types, introduction to PMBOK, portfolio Management, Traditional Vs Modern Project using PMBOK Concept **Case Study: Online Shopping**

Typical Projects Examples

- **Personal projects:**

 obtaining an MCA degree

 writing a report

 planning a party

 planting a garden

- **Industrial projects:**

 Construction of a building

 provide electricity to an industrial estate

 building a bridge

 designing a new airplane

Definition of Project

- A project is a group of unique, inter-related activities that are planned and executed in a certain sequence to create a unique product or service, within a specific time frame, budget, and the client's specifications.
- According to **Project Management Institute's** definition, a project is: "A temporary endeavor undertaken to create a unique product or service."
- The **British Standard** defines a project as: "A unique set of coordinated activities, with definite starting and finishing points, undertaken by an individual or organization to meet specific objectives within defined schedule, cost, and performance parameters."

- A project means “any undertaking that has definite, final objectives representing specified values to be used in the satisfaction of some need or desire.”
- A commercial project involves the following key considerations:

- What is the cost?
- What is the time required?
- What are the capabilities that it provides to the organization?
- Whether it will fit into the strategies of the organization?

Some of the Project Initiatives Include:

- Redesigning or relocating a production facility (Manufacturing Project)
- Implementing a management information system (MIS Project)
- Developing a new alloy required for a space vehicle (Spacecraft Project)
- Constructing national highways (Infrastructure Project)
- Organizing the Olympics (Sports Project)
- Constructing a dam for better irrigation facilities (Infrastructure Project)
- Launching of new product (Advertising and Marketing Project)
- Implementing a new computer system (IT project or upgrade)
- Designing and implementing a new organizational structure (HR Project)
- Designing and constructing a house or colony (Construction Project)

Project Parameters

Some common constraints that influence a project are:

- Scope
- Quality
- Cost
- Time
- Resources

Project Parameters: Scope

- Scope refers to *what work must be done* and *what the project will deliver*.

Why it is a constraint:

If the scope increases (more features added), the project will need:

- more time
- more cost
- more resources

Example:

- Project: Build a college attendance app

Scope includes: login, attendance, reports

Scope change: If the client adds *parent portal*, *GPS tracking*, etc., it increases workload → affecting time & cost.

Project Parameters: Quality Constraint

- Quality defines the **standard** or **level of performance** the final product must achieve.

Why it is a constraint: Higher quality needs

- better skills
- more testing
- more resources
- more time
- If time/cost is reduced, quality may suffer.

Example:

- A mobile app must load pages within 2 seconds — this is a quality requirement.
  If budget is low, developers may not be able to achieve that standard.

Project Parameters :Cost Constraint (Budget)

Cost refers to the **total amount of money** allocated to the project.

Why it is a constraint:

- The project must be completed within the approved budget.

If the budget is low:

- quality may drop
- fewer resources can be hired
- scope may have to be reduced

Example:

- College allocates ₹50,000 to develop an app.
  If the project team wants more features, the cost may cross the limit.

Project Parameters: Time Constraint (Schedule)

Time refers to the **project deadline** or duration allocated to complete all tasks.

Why it is a constraint:

- Projects must finish before the deadline.

If time is shortened:

- Scope must decrease
- More resources may be required
- Quality can suffer

Example:

- If an app must be ready before the new semester starts, the team has a strict timeline.
  If they add new features, they may miss the deadline.

Project Parameters: Resource Constraint

Resources include:

- people/team members
- equipment
- materials
- tools
- technology

Why it is a constraint:

- If resources are limited, work slows down.
  More resources increase cost.

Example:

- If only **2 developers** are available instead of 5,
  → project will take longer
  → or the scope must be reduced.

Project Parameters

Think of a project like cooking a meal:

- **Scope:** Type of meal (simple or complex?)
- **Quality:** Taste and presentation
- **Cost:** Money to buy ingredients
- **Time:** How long you have to cook
- **Resources:** Stove, utensils, ingredients, chef

If you increase scope (5 dishes instead of 2), you need more money, more time, and more people.

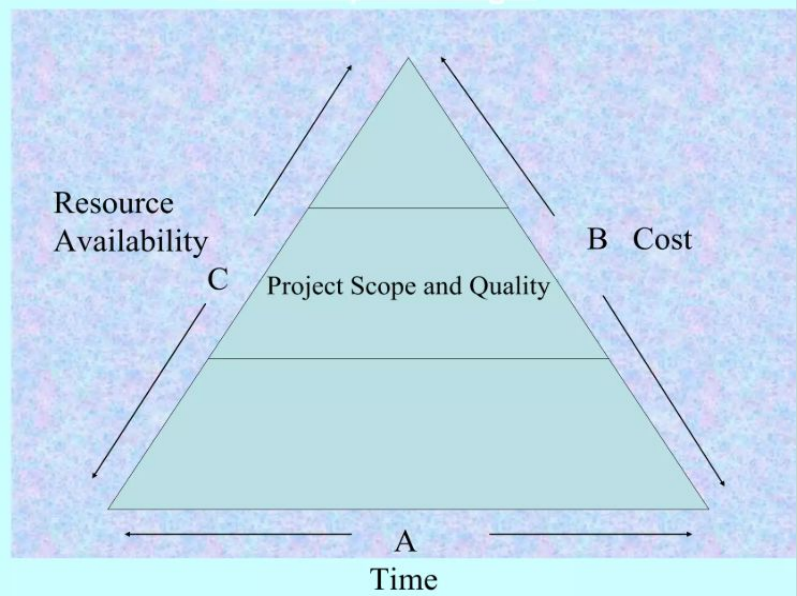
Relationship between Project Parameters

The scope and quality of a project are influenced by a variety of constraints like:

- Time
- Cost
- Availability of resources

What the Diagram Means

- If Time changes, Cost or Resources must change.
- If Cost changes, Time or Scope must change.
- If Resources change, Time or Cost will change.
- You cannot change one side without affecting the others.



Project Management Triangle

Project Management Triangle

Example: To **build a mobile app for managing assignments**.

1. Time (A)

- You want the app ready in **1 month**.

If the college says:

- "The app must launch in *2 weeks instead of 1 month*,"
  then you must change:
- **Cost** → Hire more developers (higher budget) or
- **Scope** → Reduce features (e.g., remove notification feature)

2. Cost (B)

- You have a budget of **₹50,000**.
- If the management reduces your budget to **₹30,000**, then:
- You cannot hire many developers → **Time increases**
- Or you create fewer features → **Scope decreases**

Project Management Triangle

Example: To build a mobile app for managing assignments.

3. Resource Availability (C)

- You planned **3 developers**, **1 UI designer**, **1 tester**.
- But if only **2 developers** are available:
- Work slows → **Time increases**
- To keep the deadline, you must spend more → **Cost increases**
- Or you decrease features → **Scope decreases**

Quick Real-life Example

Constraint	If it Decreases	What Happens
Time (A)	Only 1 hour instead of 3 hours	Need more bakers (Cost ↑) or produce fewer cupcakes (Scope ↓)
Cost (B)	Less money for ingredients	Buy cheaper ingredients (Quality ↓) or make fewer cupcakes (Scope ↓)
Resources (C)	Only one oven instead of two	Baking takes longer (Time ↑)

Difference between **Project** and **Program**

Project	**Program**
Project includes all activities to complete a given job.	A program is defined as an ongoing process or ongoing operation indefinitely.
It has narrow scope and has shorter duration.	It has broader scope and has longer duration.
Project is a part of a Program.	It is a main activity.
Project is an individual activity.	A program is a group of projects managed in a coordinated way to obtain benefits not available from managing them individually.

Difference between **Project** and **Program**

Project	Program
Project includes all activities to complete a given job. Ex: Developing a **college website** (only once). Activities: design pages, code backend, test, deploy When finished → Project ends	A program is defined as an ongoing process or ongoing operation indefinitely. Ex: The college has a **Digital Transformation Program**. It includes many ongoing improvements: website maintenance, app development, online admission system, data digitization No fixed end date → ongoing

Difference between **Project** and **Program**

Project	Program
It has narrow scope and has shorter duration. Ex: Organizing a **college annual day event**. Scope: stage setup, invitation design, performances Duration: a few weeks	It has broader scope and has longer duration. Ex: A **Student Skill Development Program**. Scope: multiple training workshops, guest lectures, certifications Duration: runs throughout the year

Difference between **Project** and **Program**

Project	**Program**
Project is a part of a Program.	It is a main activity.

The “Digital Learning Program” at a college may include several **projects**:

Program	Projects inside it
Digital Learning Program	1. Creating LMS (Learning Management System)portal 2. Setting up smart classrooms 3. Developing e-learning content

Difference between **Project** and **Program**

Project	Program
Project is an individual activity.	A program is a group of projects managed in a coordinated way to obtain benefits not available from managing them individually.

Program: Clean Campus Program

This program aims to improve the overall cleanliness and environmental quality.

It includes many **projects**, such as:

- Project 1: Build a waste segregation system
- Project 2: Plant 200 trees on campus
- Project 3: Install solar panels
- Project 4: Arrange student awareness workshops

Individually, each project helps.

But **combined together**, they create a healthier, greener campus → a bigger benefit.

What is Project Management?

- Managing a project is called Project Management.
- According to PMI:
  "PM is the application of knowledge, skills, tools, and techniques to project activities in order to meet or exceed stakeholder needs and expectations."
- **PM includes:**

- Defining project goals,
- Specifying how the goals will be accomplished,
- What resources are needed, and relating budgets and time for completion.

Project Management Objectives

- Coordinate the various interrelated processes of the project.
- Ensure the project includes all the work required to complete the project successfully.
- Ensure that the project is completed on time and within budget.
- Ensure that the project will satisfy the needs for which it was undertaken.
- Ensure the most effective use of the people involved with the project.
- Promote effective communication between the project's team members and key stakeholders.
- Ensure that project risks are identified, analyzed, and responded to.

Why Project Management?

The decision on whether or not to set up a separate project management division is subjective as it depends upon various factors, some of them are:

- Interactions or interdependencies between various departments.
- Sharing of common resources.
- The importance of the project to the organization.
- Size of the project.
- Degree of unfamiliarity with the work involved and its complexity.
- Changes in the market and.
- The reputation of the organization.

Project Life Cycle

- The Project Life Cycle refers to a logical sequence of activities to accomplish the project's goals or objectives.
- Regardless of scope or complexity, any project goes through a series of stages during its life.

Four Phases in the Life Cycle of a Project:

- **Phase I**: Conception and Selection
- **Phase II**: Planning and Scheduling
- **Phase III**: Implementation, Monitoring, and Control
- **Phase IV**: Evaluation and Termination

Functions of Each Phase

Phase I: Conception & Selection	Phase II: Planning & Scheduling	Phase III: Implementation, Monitoring and Control	Phase IV: Evaluation & Termination
- Identifying a need for a project	- Set up a technical team to decide on how the project can be implemented.	- Procuring materials	- Training operational staff
- Establishing goals to be achieved by the project.	- Plan for the requirements of personnel, finance, materials, etc.	- Building and testing the tools	- Transfer of materials
- Estimating the amount that the firm will have to commit for the project.	- Prepare a schedule.	- Developing support systems	- Transferring the responsibilities
- Presenting the project idea or various alternative ideas to the management and get their approval.		- Producing the system that is aimed at.	- Releasing surplus resources, that remain after use
		- Verifying whether the performance is up to the laid down standard.	- Releasing the project staff for the next assignment
		- Making modifications if required.	

Managing the Project Scope

- The scope of the project determines the boundaries of the project.
- The scope specifies what features/a characteristic of the project product is included and what is not included.
- **In project management, there are actually two different scopes:**
- **Product Scope**, which is what the end result of the project will create.
  - The product scope is what customers focus on—what they are envisioning the firm to create.
  - The product scope describes the thing or service that will exist as a result of undertaking the project.
- **Project Scope**, on the other hand, describes all the work to create the product scope.
  - It includes all of the work, and only the required work, to complete the project deliverable.

Managing the Project Scope

- Project Manager and the client prepare project scope and deliverables.
- The project manager then prepares a Project Overview Statement(PSO) based on the list of deliverables.
- The POS will be sent to the firm's top management for approval.
- The PM prepares a Project Definition Statement with his team members that can be used as a reference in executing the project.

- **Who should be the Project Manager?**
- In any organization, we find two types of people:
  - Those who give excellent ideas from their knowledge and experience, but are not very good at getting things done.
  - Those who are good at handling men and matters, but are not as sound as the former in technical matters.
- People who have great ideas are not necessarily good implementers.
- The manager of a project should be one who can work effectively with different groups of people, interact with various departmental heads, and integrate all the functions to get the project moving.
- First type of people "Project Champions“, and the second “Project Manager."

Project Manager Vs Line Managers

- There is always a constant tussle between the line managers and the project manager to share the organization's resources like:
  - Money, Manpower, Equipment, Facilities, and Information Technology.
- Manpower is controlled by Line Managers.
- Allocation of manpower will be done by Line Managers based on the request made by Project Manager.
- Employees provided to Project Manager continue to report to Line Manager and give preference to the commands of the Line Manager over the Project Manager.
- Success of the Project depends on a good working relationship between PM and LM.

Project Communication Management

Effective communication is crucial for the success of a project.

- **Project communication**, which includes both formal and informal communication at various levels, involves all the activities and behavior by which information is exchanged between the project manager and his team members.
- **Project Communication Management** includes all those processes that are required to ensure that information pertaining to the project is generated, collected, disseminated, stored, and ultimately disposed of, in a timely and appropriate manner.

Project Communication Management

- **Project communication planning** involves determining the information and communication requirements of the various stakeholders of a project.

Steps in communication planning are:

- Identifying information requirements
- Choosing the appropriate communication technology
- Stakeholder analysis
- Communication management plan

Processes and Knowledge areas in Project Management

- In **Project Management**, processes and knowledge areas are key to managing projects effectively. The **PMBOK (Project Management Body of Knowledge)**, published by the **Project Management Institute (PMI)**, outlines these processes and knowledge areas that help guide the project manager through the project life cycle.

Project Management Processes

- There are **five process groups** in project management, as defined by PMBOK:

1. Initiating Process Group
2. Planning Process Group
3. Executing Process Group
4. Monitoring and Controlling Process Group
5. Closing Process Group

Processes and Knowledge Areas in Project Management

Initiating Process Group	Planning Process Group:	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
Purpose: Defines the project and obtains authorization to start the project.	**Purpose**: Establishes the scope, objectives, and steps to be followed for the project.	**Purpose**: Implements the project plan by coordinating people and resources.	**Purpose**: Ensures the project is on track by measuring and controlling project performance.	**Purpose**: Finalizes all activities to formally close the project.
Key activities: - Develop Project Charter (Formal, high-level document) - Identify Stakeholders	**Key activities**: -Develop Project Management Plan -Define Scope -Create WBS (Work Breakdown Structure) -Define Activities -Plan Schedule, Costs, Quality, and Resources	**Key activities**: -Direct and Manage Project Work -Perform Quality Assurance -Acquire Project Team -Manage Stakeholder Engagement	**Key activities**: -Monitor and Control Project Work -Perform Integrated Change Control Control Scope, Schedule, Costs, Quality, and Risks	**Key activities**: -Close Project or Phase -Finalize contracts and deliverables

Knowledge Areas in Project Management

The **PMBOK Guide** defines **10 Knowledge Areas** that represent the essential disciplines or skill sets for managing projects. These areas are interrelated and span across the five process groups.

1. Integration Management:

- Ensures that the project is coordinated and that all elements work together.
- Key processes: Develop Project Charter, Develop Project Management Plan, Direct and Manage Project Work.

2. Scope Management:

- Defines and controls what is included and excluded in the project.
- Key processes: Plan Scope Management, Define Scope, Create WBS, Validate and Control Scope.

Knowledge Areas in Project Management

3. Time Management:

- Ensures that the project is completed on time by planning and controlling the schedule.
- Key processes: Plan Schedule Management, Define Activities, Sequence Activities, Estimate and Control Durations.

4. Cost Management:

- Ensures that the project is completed within the approved budget.
- Key processes: Plan Cost Management, Estimate Costs, Determine Budget, Control Costs.

5. **Quality Management**:

- Ensures that the project meets the defined quality standards.
- Key processes: Plan Quality Management, Perform Quality Assurance, Control Quality.

Knowledge Areas in Project Management

6. **Resource Management**:

- Ensures effective use of resources (human, material, etc.) to achieve project goals.
- Key processes: Plan Resource Management, Estimate Resources, Acquire and Develop Team, Manage Team, Control Resources.

7. **Communications Management**:

- Ensures timely and appropriate generation, collection, distribution, storage, and ultimate disposition of project information.
- Key processes: Plan Communications Management, Manage Communications, Monitor Communications.

8. **Risk Management**:

- Identifies, analyzes, and responds to project risks to minimize negative impacts.
- Key processes: Plan Risk Management, Identify Risks, Perform Qualitative and Quantitative Risk Analysis, Plan and Implement Risk Responses.

Knowledge Areas in Project Management

9. Procurement Management:

- Ensures the project obtains the necessary resources and services from external sources.
- Key processes: Plan Procurement Management, Conduct Procurements, Control Procurements, Close Procurements.

10. Stakeholder Management:

- Identifies and manages stakeholders to ensure their expectations are met, and their influence on the project is properly managed.
- Key processes: Identify Stakeholders, Plan Stakeholder Engagement, Manage Stakeholder Engagement, Monitor Stakeholder Engagement.

Work Breakdown Structure (WBS)

- The **Work Breakdown Structure (WBS)** is a **fundamental project management tool** that helps break down a project into smaller, more **manageable components**. It provides a clear hierarchical structure, making it easier to **understand the project scope, organize the tasks, assign responsibilities, and track progress**.
- A WBS is a **deliverable-oriented breakdown** of the work required to complete the project. It organizes the project into **work packages, which are the smallest units of work that can be assigned, estimated, and tracked.**

Work Breakdown Structure (WBS)

Purpose of WBS

Scope Definition: It helps define the scope of the project and ensures that all the work required to complete the project is identified.

Project Planning: Facilitates detailed planning by breaking down large tasks into smaller tasks.

Resource Management: Helps in assigning resources, managing budgets, and tracking progress.

Risk Management: Identifies potential risks by reviewing the work breakdown structure.

Improved Communication: Provides a clear view of the project structure for all stakeholders.

Structure of a WBS

A WBS is typically represented as a **tree diagram**, where:

- The **top level** represents the **overall project**.
- The **subsequent levels** represent **major deliverables, components, and smaller tasks until reaching work packages**.
- Each work package is further divided into **smaller tasks** that can be easily **managed and tracked**.

Types of WBS

There are several types of WBS that can be used depending on the nature of the project and its deliverables.

1) Deliverable-Based WBS

- **Definition**: A **Deliverable-Based WBS** is organized around the key deliverables or products of the project. It focuses on the **outcomes** that the project will produce.
- **Example**:
  In a **software development project**, the WBS might be organized by key deliverables, such as:
  - **Requirements Gathering**
  - **System Design**
  - **Development (coding)**
  - **Testing**
  - **Deployment**
- **Benefits**: This approach is ideal for projects with **well-defined end results or outputs (products or services)** and works well for **ensuring all deliverables** are completed.

Types of WBS

2. Phase-Based WBS

- **Definition**: A **Phase-Based WBS** organizes the project **according to its phases or stages**. It divides the project into **sequential phases**, which are further broken down into **smaller tasks or activities**.
- **Example**:
  In a **construction project**, the WBS might follow phases like:
  - Planning Phase
  - Design Phase
  - Construction Phase
  - Testing/Inspection Phase
- **Benefits**: This approach is ideal for projects that have **clear stages or steps**, such as construction, product development, or event planning. It is useful for projects where the **scope is executed in distinct, ordered phases.**

Types of WBS

3. Hybrid WBS

- **Definition**: A **Hybrid WBS** combines both **deliverable-based** and **phase-based** structures. It organizes the project by both the deliverables and the phases in which they are created.
- **Example**:
  In an **IT infrastructure project**, the WBS might be:
  - Phase 1: Requirements Analysis
    - Deliverable: Requirements Document
  - Phase 2: System Design
    - Deliverable: Design Document
  - Phase 3: Implementation
    - Deliverable: Fully Functional System
- **Benefits**: This approach is useful when a project requires both phase-specific activities and a focus on deliverables. It allows the project manager to track progress and ensure all deliverables are completed at each stage.

Types of WBS

4. Organizational WBS

Definition: An **Organizational WBS** is based on the **structure of the organization or departments** that will be responsible for different parts of the project. This type of WBS focuses on the **responsibilities of each organizational unit.**

Example:
In a **large corporate project**, the WBS might be organized by departments such as:

- IT Department
  - Develop System Architecture
  - Implement Software Solutions
- HR Department
  - Recruit Staff
  - Employee Training

Benefits: This approach is useful for large projects with multiple departments or teams. It provides clarity on which department is responsible for which deliverables or tasks.

Types of WBS

5. Product-Based WBS

- **Definition**: A **Product-Based WBS** focuses on **the features or characteristics of the final product or service being developed**. Each work package represents a specific feature or component of the product.
- **Example**:
  In an **automobile manufacturing project**, the WBS might include:
  - Engine Design
  - Body Design
  - Assembly Process
  - Quality Control
- **Benefits**: This approach is best for product development projects, as it ensures all aspects of the product are covered, focusing on components or features.

WBS Levels

- A typical WBS has multiple levels, with each subsequent level providing more detail:
- **Level 1**: The **overall project or program.**
- **Level 2**: Major **deliverables or project phases.**
- **Level 3**: **Work components** within deliverables or phases.
- **Level 4**: **Specific tasks or work packages**.

Example of WBS (for a Web Development Project)

Level 1 (Overall project or program)	Level 2 (Deliverables or project phases)	Level 3 (Work components in project)	Level 4 (Specific tasks)
Web Development	**Planning**	Define Requirements	Meet with Stakeholders
		Define Scope	Document Functional Specifications
	Design	Create Wireframes	Design Home Page
		Create Prototypes	Design Product Pages
	Development	Frontend Development	Write HTML/CSS for Homepage
			Implement User Login System
	Testing	Perform Unit Testing	Test Homepage Responsiveness

Benefits of WBS:

- **Clear Structure:** WBS provides a visual and organized way to break down the project into **manageable chunks, making it easier to assign tasks and monitor progress**.
- **Scope Management:** It ensures that all work required to complete the project is identified and assigned, preventing scope creep by **clearly defining what is included and excluded**.
- **Better Resource Allocation:** With a clear understanding of the work involved, the project manager can **allocate resources** (human, financial, material) more effectively.

Benefits of WBS:

- **Improved Communication**:
  - The WBS serves as a common reference point for the team, helping everyone understand their responsibilities and the project's overall structure.
- **Risk Identification**:
  - By breaking down the project into smaller tasks, the project manager can better identify potential risks and prepare mitigation strategies.

Introduction to PMBOK (Project Management Body of Knowledge)

- set of guidelines, best practices, and standards for project management, developed and maintained by the Project Management Institute (PMI).
- The PMBOK serves as a globally recognized standard for managing projects and offers a structured framework that ensures projects are executed successfully, on time, and within scope and budget.
- It gives a framework and common language so project managers can plan, execute, monitor, and close projects in a structured way

Introduction to PMBOK (Project Management Body of Knowledge)

What is PMBOK?

- **PMBOK** is a guide that **describes the processes, knowledge areas, tools, and techniques** essential for managing a project. It is used by project managers worldwide to **ensure consistency and success** in project management.
- It is organized into **processes and knowledge areas**, providing a structured approach to managing projects.
- The **PMBOK Guide** is continuously updated to reflect the latest trends and practices in project management, and it is essential reading for anyone pursuing project management certification, such as the **Project Management Professional (PMP)** exam.

Portfolio Management

- **Portfolio Management** refers to the centralized management of one or more portfolios, which are collections of projects and programs that are grouped to facilitate effective management and meet strategic objectives. Portfolio management is focused on maximizing the value and benefits of all projects and programs in alignment with an organization's overall goals.
- The primary goal of **portfolio management** is to ensure that the projects and programs being executed are aligned with the organization’s strategy, and that the right projects are selected, prioritized, and managed in a way that optimizes resources and minimizes risks.

Key Concepts of Portfolio Management

Portfolio	Program	Project
A **portfolio** is a collection of projects and programs that are managed together to achieve strategic objectives. It includes all projects within an organization, regardless of their size, scope, or phase of development. Example: A software company's portfolio could include a mobile app development project, a new software product launch, and an IT infrastructure upgrade.	A **program** is a group of related projects that are managed in a coordinated way to obtain benefits and control not available from managing them individually. Example: A company's program could include a series of software development projects that are all aimed at improving customer experience.	A **project** is a temporary endeavor undertaken to create a unique product, service, or result. Example: Developing a specific mobile app for a particular market.

Importance of Portfolio Management

Importance of Portfolio Management

Strategic Alignment

Example: If a company's strategic objective is to expand into new markets, portfolio management helps identify and prioritize projects related to international expansion.

Resource Optimization

Example : If a company has only a few software developers, portfolio management helps decide **which projects are most important** so that those developers work on the projects that matter most to the company.

Risk Management

Example: If one project is delayed or runs out of resources, it may slow down other projects that depend on the same team or budget. Portfolio management helps the company **see this connection** and **fix the issue** before it spreads.

Increased Efficiency

Example: Two software development projects working on similar technologies can share resources, reducing costs and effort.

Decision Making

Example: If one project is not performing well and is wasting money or time, portfolio management helps the company decide to **stop that project** and shift those resources to another project that is **doing well and has better future results**.

Portfolio Management Process

Portfolio Planning	Portfolio Selection	Portfolio Execution	Portfolio Monitoring and Control	Portfolio Review And Evaluation
-Identify business objectives. -Understand available resources -Decide the types of projects needed	-Organization chooses the best projects from many possible options. -Compare project ideas -Select projects that match goals -Check feasibility and benefits	- Assign teams and resources - Implement project plans - Coordinate tasks across projects	- Track schedule, cost, and quality - Identify risks and issues - Adjust resources as needed	-Measure performance -Review benefits achieved -Learn lessons for future portfolios

Portfolio Management Process

Portfolio Planning

A hospital wants to improve patient care. During portfolio planning, it decides to start projects such as:

- Implementing an online appointment system
- Setting up a training program for nurses
- Upgrading medical equipment in critical departments

This helps the hospital choose the right projects that support its overall goal of better patient care.

Portfolio Selection

A school receives several project proposals to improve student learning. After reviewing them, the school selects:

- **Installing a digital library** (easy to start and highly beneficial)
- **Launching a student skill development program** (matches the school's goals)

It rejects other proposals that are too costly or not useful enough.

This shows how the organization picks the **most valuable and achievable projects** from many options.

Portfolio Management Process

Portfolio Execution

Imagine a company has approved a portfolio that includes:

- Developing a **mobile application**
- Launching a **digital marketing campaign**
- Upgrading the **IT security system**

During Portfolio Execution:

- The **mobile app team** begins working on app design, user interface, and coding.
- The **marketing team** starts creating social media posts, videos, and promotional content.
- The **IT team** starts installing new security tools and updating internal systems.

Portfolio Management Process

Portfolio Monitoring and Control

Example:

A company is running several projects in its portfolio:

- Mobile app development
- IT security upgrade
- Digital marketing campaign

During **Portfolio Monitoring and Control**:

- Managers review **monthly progress reports** from each project team.
- They compare actual progress with the **planned schedule and budget**.
- If the mobile app project is running late, they may **add more developers** or **extend deadlines**.
- If the marketing campaign is using more budget than planned, they might **reduce low-priority activities**.
- If the IT security upgrade finishes early, those resources may be **reassigned** to help other projects.

Portfolio Management Process

Portfolio Review And Evaluation

Example:

A company has completed several projects in its portfolio:

- Mobile app development
- Digital marketing campaign
- IT security upgrade

During **Portfolio Review and Evaluation**:

- Senior managers examine whether each project met its **objectives**, such as increasing customer engagement or improving system security.
- They review **key performance indicators (KPIs)** like cost, time, quality, and customer satisfaction.
- If the mobile app increased user sign-ups by 40%, it is marked as highly successful.
- If the marketing campaign did not meet its expected reach, they analyze **what went wrong** and identify improvements for future campaigns.
- Lessons learned from all projects are documented and used to update **portfolio strategies** and decision-making processes.

Types of Portfolio Management

- Strategic Portfolio Management
- Operational Portfolio Management
- Financial Portfolio Management

Strategic Portfolio Management (SPM)

- **Strategic Portfolio Management** is the systematic process of selecting, prioritizing, and managing an organization's projects, programs, and initiatives so that they are aligned with its long-term goals and strategic direction.
- The aim is to ensure that every project contributes measurable value, resources are optimally utilized, and risks are controlled at the organizational level.

Strategic Portfolio Management (SPM)

SPM helps leadership answer key questions such as:

- *Are we investing in the right projects?*
- *Do these projects support our organizational mission and strategy?*
- *How should we allocate budgets, people, and technology across all initiatives?*
- *Which projects should be started, continued, paused, or stopped?*

Operational Portfolio Management

- **Operational Portfolio Management (OPM)** focuses on managing and monitoring the set of ongoing, day-to-day projects, services, and activities that support an organization's **routine operations**.
- While *Strategic Portfolio Management* ensures alignment with long-term goals, **Operational Portfolio Management ensures that current operations run efficiently, consistently, and within available resources**.

Operational Portfolio Management

It includes:

- Managing ongoing projects and tasks
- Ensuring operational efficiency
- Tracking performance indicators (KPIs)
- Managing risks, costs, and resources
- Coordinating people, processes, and services
- Ensuring continuous improvement in operations
- The primary goal is to **deliver reliable, stable, and high-quality operations every day**.

Financial Portfolio Management

Financial Portfolio Management refers to the process of selecting, organizing, and managing a collection of financial assets—such as stocks, bonds, mutual funds, real estate, and cash—in a way that balances **risk, return, and investment goals**.

It ensures that money is invested in the right mix of assets, based on:

- Expected returns
- Risk appetite
- Time horizon
- Liquidity needs
- Market conditions

The aim is to **maximize returns while minimizing risks** by diversifying investments and making informed financial decisions.

Financial Portfolio Management

It includes:

- Setting financial goals
- Asset allocation (how much to invest in each asset type)
- Risk assessment and balancing
- Diversification
- Continuous monitoring and rebalancing
- Reviewing market performance

Tools and Techniques in Portfolio Management

1. **Portfolio Matrix**: A graphical tool used to evaluate and prioritize projects based on factors such as risk, return on investment (ROI), strategic alignment, and resource requirements.
2. **Financial Metrics:** Tools such as **Net Present Value (NPV)**, **Return on Investment (ROI)**, and **Internal Rate of Return (IRR)** are often used to assess the financial viability of projects within a portfolio.
3. **Risk Assessment**: Portfolio management tools often include risk assessments to evaluate the risk exposure of each project and the overall portfolio. This helps in balancing the portfolio with a mix of high-risk and low-risk projects.
4. **Balanced Scorecard**: A strategic planning tool used to monitor the performance of the portfolio. It includes multiple perspectives such as financial, customer, internal processes, and learning & growth.

Traditional Vs Modern Project using PMBOK Concept

PMBOK recognizes that **projects can be managed using different approaches**. The two major ones are:

- **Traditional / Predictive Approach** — “Plan first, then execute”
- **Modern / Agile–Adaptive Approach** — “Adapt as you go”

Traditional Vs Modern Project using PMBOK Concept

Traditional (Predictive) Project	Modern (Agile/Adaptive) Project
-You plan everything in advance and follow the plan strictly. -Best when requirements are clear and stable Ex: Building a bridge Road construction Manufacturing plant setup	-Work is done in small cycles (iterations) and the plan keeps improving based on feedback. -Best when requirements keep changing Ex: Software development App creation Research & innovation projects
Traditional Approach aligns with: -Initiating -Planning (very detailed plan) -Executing -Monitoring & Controlling -Closing **Heavy focus on Planning first**	**Modern (Agile/Hybrid) Approach aligns with:** -Repeated cycles of **Plan → Do → Review → Improve** -Continuous stakeholder involvement -Flexible planning **Planning + Execution happen together in short cycles**